

Hugo Cisneros

Lead ML Engineer with PhD background in AI, specializing in production ML systems and data platforms.

Website: <https://hugocisneros.com>

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Google Scholar: [Scholar Profile](#)

WORK & RESEARCH EXPERIENCE

- Mar 2023 - *Current* **Inicio (Startup) Lead Machine Learning Engineer**, Paris
- **Led core logic team** (5 engineers + QA) through company growth from **5-person startup to 35 employees** post-seed funding; drove technical direction for data engineering, ML/AI modeling, and geoprocessing infrastructure
 - **Scaled renewable energy site selection platform** to process **entire countries in <24h** (from 1h/city), scanning **~10 TB of geospatial data** with **100+ simultaneous constraints** using distributed AWS infrastructure (ECS, Aurora PostgreSQL/PostGIS)
 - Built **end-to-end ML pipelines**: computer vision for urbanism map segmentation (**SAM - Segment Anything Model**, classical CV); **LLM-based document parsing** and knowledge graph extraction; project success prediction models. Tech stack: Python, Rust, PyTorch, AWS
 - Developed **grid intelligence platform** extracting structured data from **thousands of regulatory PDFs**, creating comprehensive map of solar projects in development across **France, Italy, and UK** for **~100 enterprise users**
 - Mentored junior engineers and interns on ML and software engineering best practices, code review, and system design

Completed PhD (2019-2023, see Education below)

- Apr - Nov 2019 **CIIRC (Czech Institute of Informatics, Robotics and Cybernetics) Research Intern**, Prague
- Under the supervision of Tomas Mikolov (Facebook AI Research).
- Studied emergence, complexity, and spontaneous organization in complex systems and their applications to artificial intelligence.
 - Built a neural network-based complexity metric for measuring emergence in cellular automata and other dynamical systems (implemented in C) which led to a peer-reviewed publication. [\[GitHub\]](#)

- Mar - Sep 2018 **INRIA and CNRS (LIMSI) Research Intern**, Paris
- Under the supervision of Xavier Tannier and Ioana Manolescu.
- Built a pipeline generator for extracting and integrating multiple data sources with **Natural language processing (NLP) and data processing algorithms** for data journalism. [\[GitHub\]](#)
 - Collaborated with journalists from *Le Monde* ("*Les Décodeurs*") on automating their data processing pipelines and using NLP for their investigations.
 - Reviewed literature on **machine learning in graphs, automatic knowledge base construction** and **natural language processing for fact checking**.

- Jun 2017 - Mar 2018 **Aiden.ai Software Engineering and Machine Learning Intern**, London
- Built an **AI powered virtual colleague** for Marketing analysts based on **Natural Language Processing** intent recognition (LUIS)
 - Implemented **machine learning pipelines** with Python for marketing data forecasting, classification and user clustering (scikit-learn, Keras, FB Prophet)
 - Participated in implementing the chat interface and the Natural Language recognition system with **Typescript**

- Sep 2016 - Feb 2017 **ENS Ulm, Kastler-Brossel Laboratory Research Assistant**, Paris
- Light control and propagation in amplified multimode fibers
- Implemented and optimized **finite elements simulations** with Python and Matlab
 - Performed **high performance computing** on large distributed clusters
 - Worked with PhD candidate on building a tool for optimizing the propagation of a light beam in optical fibers [\[Report\]](#)

EDUCATION

- May 2023 - Nov 2019 PhD, **INRIA, CIIRC CTU** (Czech Technical University in Prague), Paris & Prague
- Unsupervised learning with Complex Systems and Evolution***

Under the supervision of Tomas Mikolov and Josef Sivic. Topics: complex dynamical systems, self-organization, artificial evolution, artificial intelligence.

- Developed and trained **deep learning models** (PyTorch, custom architectures) for studying emergence and complexity in dynamical systems
- Built **highly optimized C code** for large-scale simulations of cellular automata and complex systems
- Conducted **distributed computing** experiments using **SLURM clusters** and **OpenMPI** for parallel processing
- Published **4 peer-reviewed papers** in conferences and journals (CoLLAs, ALIFE, IEEE SSCI)
- Supervised Master-level theses and internship projects

Sep 2019 - Sep 2018	MVA Master in Machine Learning and Applied Mathematics, ENS Paris Saclay , Paris Relevant Coursework: Convex Optimization, Probabilistic Graphical Models, Computer Vision, Reinforcement Learning, Deep Learning, Speech and Natural language processing, Kernel Methods, Biostatistics, Theoretical Foundations of Deep Learning — (GPA: 16.2 / 20)
Sep 2018 - Sep 2015	Master of Science in Engineering, Mines ParisTech , Paris Specialization: Computer Science — (3.7 GPA) Relevant Coursework: Machine Learning, Probabilities, Statistics, Programming
Aug 2015 - Sep 2013	Preparatory class for <i>Grandes Ecoles</i> Lycée Stanislas (Paris) MPSI and MP* Bachelor's Degree in Mathematics and Physics, national competitive exam for entering engineering school.
Aug 2013	Scientific Baccalauréat (High school diploma in Maths, Physics and Life Sciences) - High distinction

PUBLICATIONS

Cisneros, H. **Unsupervised Learning in Complex Systems**. Thesis.

Link: <https://arxiv.org/abs/2307.10993>

Herel, D., Cisneros, H., & Mikolov, T. **Preserving Semantics in Textual Adversarial Attacks**. Preprint.

GitHub repo: <https://github.com/DavidHerel/semantics-preserving-encoder>

Cisneros, H., Sivic, J. & Mikolov, T. **Benchmarking Learning Efficiency in Deep Reservoir Computing**. First Conference on Lifelong Learning Agents (CoLLAs 2022).

GitHub repo: https://github.com/hugcis/benchmark_learning_efficiency

Cisneros, H., Sivic, J. & Mikolov, T. **Visualizing computation in large-scale cellular automata**. Artificial Life Conference Proceedings 32, 239–247 (2020).

Cisneros, H., Sivic, J. & Mikolov, T. **Evolving Structures in Complex Systems**. in 2019 IEEE Symposium Series on Computational Intelligence (SSCI) 230–237 (IEEE, 2019).

GitHub repo: <https://github.com/hugcis/evolving-structures-in-complex-systems>.

PROJECTS

Mar - Aug 2021	Participated in the Open-endedness evolution challenge at the GECCO 2021 conference competition track. Developed an open-ended algorithm based on Neural Cellular Automata in Pytorch within the game Minecraft. Finished second place. [GitHub][Blog post]
Jun - Aug 2018	Participated in the n2c2 shared task of Harvard Medical School <i>Cohort Selection for Clinical Trials</i> in a joint team from AP-HP and LIMSI. Implemented weakly-supervised and transfer learning methods for Medical NLP (Keras). Finished 2nd among 30 teams. [Preprint]
Jan 2018	Built a NLP based tool for discovering and matching similar arXiv papers based on similarity measures including word embeddings-based similarities of their abstract and co-authorship graph distance . [GitHub]

Feb 2017 Implemented a multi-currency blockchain in Python with a team of 9 people (Cryptography, network programming, team software development)

PROGRAMMING SKILLS

ML/AI: PyTorch, Langchain, TensorFlow, Keras
Languages: Python, C, Rust, Java, TypeScript, Matlab, Scala, Ruby, C++
Data: SQL (PostgreSQL), Airflow, Dagster
Tools & Platforms: Git, Docker, AWS (ECS, S3, Aurora), Terraform, LaTeX

LANGUAGES

ENGLISH: Fluent	SPANISH: Intermediate
FRENCH: Native	JAPANESE: School level

INTERESTS AND ACTIVITIES

- Creative coding, Live Coding, Generative Art, <https://genocto.xyz>
- Technology, Open-Source, Programming
- Mathematics, Statistics and Probabilities
- Running, Hiking, Fencing, Piano, Guitar